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The role of wildlife in transboundary animal diseases

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Abstract

This paper identifies some of the more important diseases at the wildlife–livestock interface and the role wildlife plays in disease transmission. Domestic livestock, wildlife and humans share many similar pathogens. Pathogens of wild or domestic animal origin that can cause infections in humans are known as zoonotic organisms and the converse are termed as anthroponotic organisms. Seventy-seven percent of livestock pathogens and 91% of domestic carnivore pathogens are known to infect multiple hosts, including wildlife. Understanding this group of pathogens is critical to public health safety, because they infect a wide range of hosts and are most likely to emerge as novel causes of infection in humans and domestic animals. Diseases at the wildlife–livestock interface, particularly those that are zoonotic, must be an area of focus for public health programs and surveillance for emerging infectious diseases. Additionally, understanding wildlife and their role is a vital part of understanding the epidemiology and ecology of diseases. To do this, a multi-faceted approach combining capacity building and training, wildlife disease surveillance, wildlife–livestock interface and disease ecology studies, data and information sharing and outbreak investigation are needed.

Keywords: wildlife–livestock interface, disease, pathogen, anthroponoses, zoonoses, wildlife, livestock

Introduction

Domestic livestock and humans share many similar pathogens (Haydon *et al*., 2002). Pathogens that originate in domestic or wild animals and cause infections in humans are known as zoonotic organisms; pathogens that originate with humans and cause disease in animals are termed as anthroponotic organisms. Zoonotic diseases account for about 61% of all the pathogens infecting humans and therefore are an important public-health issue worldwide. Furthermore, as the global human population continues to increase, anthroponosis is now an emerging issue. Some human diseases currently pose conservation risks to wildlife (Taylor *et al*., 2001). A high percentage (77%) of livestock pathogens and an even higher percentage (91%) of domestic carnivore pathogens infect multiple hosts, including wildlife (Cleaveland *et al*., 2001). Understanding this group of pathogens is critical to public health, because infection of a wide range of hosts increases the likelihood that these organisms will emerge as novel infections in humans and domestic animals. Therefore, diseases at the wildlife–livestock interface, particularly those that are zoonotic, are of paramount importance and must be an area of focus for public health programs and surveillance for emerging infectious diseases.

For the diseases that wildlife and livestock share, examination of the factors affecting disease ecology is essential. For infection to occur in a host, the animal must accommodate a pathogen in its body, exploit host cells and evade the natural defense mechanisms of the host

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(Hudson, 2002). To support the pathogen cycle that allows for effective transmission of the pathogen, a maintenance host is needed. Commonly, the pathogen will establish infection in the maintenance host and induce clinical disease, leading to maximum pathogen transmission. Another key ecological factor is the role of a reservoir host. Conventionally, a reservoir host can asymptomatically maintain a pathogenic organism and transmit it to other animals where disease occurs. A species' population can also act as a reservoir as long as the reproductive number (R_0) of the infection exceeds 1; this occurs when a population is of sufficient size and enough susceptible animals are present over time to propagate infection (Haydon *et al.*, 2002; Hudson, 2002).

Occasionally, wildlife is the original source and maintenance host of the organism, and therefore the source of infection for livestock. Conversely, livestock can introduce disease to wildlife and although the wildlife species may not be a competent maintenance host, occasionally they are able to transmit the infection within and between species for a period of time; this is referred to as a spill-over host or bridge species (Daniels *et al.*, 2007). A third situation involves dead-end hosts, infected animals in which the organism loses virulence over time and/or the animal dies; thus, the organism cannot be transmitted (Hudson, 2002). Disease ecology is extremely complex, and in many cases the reservoirs and dead-end hosts have not been identified (Haydon *et al.*, 2002).

The interface of wildlife and livestock is multi-dimensional. The two most critical factors are the location and time period in which diseases are exchanged at this interface. These interfaces are vast and include both direct and indirect contact. Commonly, the interface is at physical contact points where both populations access communal resources, such as rangeland, watering and feeding sites. This makes it difficult to prevent interaction between species without considerable spatial separation and infrastructure investment. Direct contact occurs through contaminated aerosols, feed, water or environment, while indirect contact occurs with infected flightless vectors, winged vectors or mechanical vectors. In situations where a natural barrier exists or where the contagion is released post mortem, wind, fomites, winged and mechanical vectors play a more important role in disease transmission (Bengis *et al.*, 2002; Hudson, 2002).

Climatic events also affect disease transmission by shifting host and vector distributions, altering pathogen survival and impacting nutritional planes and immunities in a population during extreme climate conditions. As an example, the prevalence of Rift valley fever, a vector-borne disease, is dependent on precipitation that drives the emergence of mosquito populations. Conversely, diseases that are transmitted by direct contact (such as foot and mouth disease (FMD) and rinderpest) have a higher prevalence during dry seasons, when a dearth of water increases animal densities near watering holes. Climatic events also affect pathogen survival with cool and humid environments allowing pathogens to survive longer. Severe weather events can affect host immunity and nutrition if they interrupt forage availability and feeding patterns.

Undoubtedly, one of the most important factors driving disease emergence is the establishment of new interfaces between wildlife and livestock through intensification of farming, altering landscapes and demographics, regional conflict, political instability, irresponsible trade, animal translocations and/or animal or plant introductions. These mechanisms enhance disease transmission as a result of changing the spatial distribution of hosts and increasing contact rates between animal populations (Bengis *et al.*, 2002; Kruse *et al.*, 2004). Where interspecies transmission is novel, it is described as pathogen jumping or host switching. This interspecies progression increases opportunity for pathogen evolution through a variety of mechanisms, e.g. genetic reassortment and recombination, occurring when different strains of the same microorganism are mixed together in one host (Vande-Woude and Apetrei, 2006; Parrish *et al.*, 2008).

Infectious diseases of livestock can cause major losses in trade income in livestock-producing countries in developing regions through denying access to profitable markets and threatening food security and livelihoods worldwide. The infectious diseases that are highly contagious and spread rapidly regardless of country borders are referred to as transboundary animal diseases (TADs) (Baldock *et al.*, 1999). In this review, we describe the important TADs at the wildlife–livestock interface. Diseases of livestock (including swine, poultry, large and small ruminants) in which wildlife is potentially involved in disease transmission cycles were included on the basis of being highly contagious and spreading rapidly across borders, causing major loss in livestock production through trade embargos or livestock mortality and morbidity.

Ruminant diseases

Viral diseases

Foot and mouth disease virus (FMDV), is highly contagious and infects cloven-hooved animals, causing vesicular lesions associated with the oral cavity, coronary bands of the hoof, interdigital skin and the udder. Adult animals generally recover from infection but naïve and young animals suffer high mortality (Thomson and Bastos, 2004). Undiagnosed disease can cause major constraints in international livestock trade and result in embargos by trade partners on livestock and livestock products (FAO, 2008). In wildlife species, under natural ecological conditions, FMD is rarely noticed but with altered landscapes and isolated fenced populations, introduction of virus can result in significant mortality

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and disease as seen in the case of impala (*Aepyceros melampus*) (Keet *et al.*, 1996).

There are seven immunologically distinct serotypes of FMDV. The serotypes and strains (over 60) vary with each geographic region, with serotype O being the most common worldwide. Cross-protection between strains can occur depending on their antigenic similarity (Davies, 2002). Reservoir hosts are numerous and include domestic ruminants and pigs, and among wildlife, African buffalo (*Syncerus caffer*), reindeer (*Rangifer tarandus tarandus*) and wild boar (*Sus scrofa*) (Thomson *et al.*, 2003; Hargreaves *et al.*, 2004; Thomson and Bastos, 2004; Jori *et al.*, 2009). Furthermore, 60–70 species of wild animals have been reported to be susceptible to natural or experimental infection with FMDV, and on several occasions FMDV was isolated from some in the natural state (Pinto, 2004).

Wherever in the world FMDV has been eliminated from livestock, it also disappeared from wildlife in those regions (Thomson and Bastos, 2004). This is most likely due to insufficient population sizes of susceptible wildlife to maintain the virus in these situations. The exception is in parts of sub-Saharan Africa where African buffaloes serve as a reservoir for the SAT-type viruses. Serological surveillance in wildlife including the use of liquid-phase blocking enzyme-linked immunosorbent assay (*Ceditest*®, Prionics AG, Zurich) has shown these viruses to be widespread among African buffalo, impala and kudu (*Tragelaphus* spp.) across southern and eastern Africa (Hargreaves *et al.*, 2004; Bronsvoort *et al.*, 2008; Jori *et al.*, 2009) and gazelles in Mongolia (Nyamsuren *et al.*, 2006). In the last three outbreaks with major economic losses in Taiwan, United Kingdom and Argentina, wildlife were not associated with the introduction of disease (Ward *et al.*, 2007), and it is likely that wildlife-to-livestock infection is a rare event even in endemic settings.

Transmission occurs through direct or indirect contact with infected animals or contaminated fomites e.g. vehicle tires. FMDV is found in all excretions and secretions (saliva, milk, urine, feces and semen) from acutely infected animals and is easily aerosolized (Thomson and Bastos, 2004). Shedding of the virus varies by species with infected *Bovidae* shedding the virus for six months and smaller hoofstock from four to nine months. African buffalos acquire infection within a few months of life and can shed the virus for up to five years. The virus can persist in a herd for up to 24 years, and sexual transmission has been reported in buffalos (Pinto, 2004).

A historic approach in southern Africa to minimize transmission between wildlife reservoirs and livestock is the physical separation of the two groups with tall reinforced fencing and FMDV vaccination of cattle in localities adjacent to wildlife areas (Hargreaves *et al.*, 2004). This large-scale intervention has generally worked in reducing the prevalence of FMDV. However, the use of fencing is controversial because fences may block migration routes, access to water and other resources needed for wildlife. The alternative to fencing is relying on vaccination exclusively; however, the virus has many antigenic variations often rendering vaccination ineffectual where multiple virus types frequently circulate (Ward *et al.*, 2007). Vaccination requires several inoculations to reach the required level of immunity and yearly vaccination is required to sustain protective immunity. Commodity-based trade, incorporating meat processing, rendering the product of negligible risk for FMDV (removal of lymph nodes and bones) shows considerable promise of opening up trade for areas of Africa where landscape precludes conventional control options (Thomson *et al.*, 2004).

Another important disease (although almost eradicated) is rinderpest, an acute, highly contagious, viral disease of cattle, domesticated buffalo and some species of wild artiodactyls. After the development of safe effective vaccines (Plowright, 1968) and earlier failed attempts at virus eradication at the continental level (JP15), in 1992, the Food and Agriculture Organization of the United Nations (FAO) started the Global Rinderpest Eradication Programme with the goal of complete eradication by the year 2011. If successful, this would be the first worldwide eradication of an animal pathogen (Barrett *et al.*, 2006). There is only one serotype of the virus but there are three genetically distinct lineages (1, 2 and 3). Cattle, sheep, goats and yaks are susceptible and any cloven-hooved animal, domestic or wild can become infected (Roeder and Taylor, 2002). African buffalo and other species are biologically competent to be reservoirs; but the belief is that wildlife cannot maintain the virus without a cattle source due to their relatively low population numbers; hence, they are not acting as a true reservoir (Kock *et al.*, 1999). However, wildlife is still important as vectors and where the virus is present, buffalos play a critical role in epidemics, spreading the virus across considerable distances over time (Kock *et al.*, 2006).

Transmission is by direct or close indirect contact with infected animals via aerosol transmission. Transmission by fomites is possible; however, the virus is readily inactivated by sunlight and drying (Barrett *et al.*, 2006). The disease history is one of devastating epizootics with the most significant being in Africa toward the end of the 19th Century (Spinage, 2003). The persistence of rinderpest in the absence of animal movement control and vaccination is remarkable. After centuries of adaptation, the virus strains evolved and many milder cases were recorded. This change complicated the final phase of eradication of the virus from Africa, and the last remaining strain was mild in cattle and cryptic.

Fortunately, with only cryptic forms of infection in cattle persisting in the 21st century, its expression in wildlife, which was more often severe, enabled rinderpest surveillance to identify remaining foci (Kock, 2008).

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As the virus is most often introduced by infected animals, the control of outbreaks includes quarantine and movement control, euthanasia of infected and exposed animals, decontamination of infected premises and focal vaccination (Thomson and Bastos, 2004).

Like rinderpest, Peste des petits ruminants (PPR) is caused by a morbillivirus. It is a highly contagious viral disease, mainly in sheep and goats which are important livestock species for many of the world's poor (Barrett *et al.*, 2006). PPR is also an emerging disease of semi-captive wildlife (Abubakar *et al.*, 2009) and recently free-ranging caprines in Pakistan (Qadir Shah Ghulam, personal communication 2009), buffalo in India (Govindarajan *et al.*, 1997) and in captive and semi-captive gazelles (*Gazelle dorcas* and *Gazelle thomsoni*) in Saudi Arabia (Elzein *et al.*, 2004). The virus has continued to spread throughout Africa, the Middle East and Asia since its discovery in 1942 in West Africa (Abubakar *et al.*, 2009). Antibodies to PPR and rinderpest are cross-protective and vaccination for rinderpest can mask the presence of PPR (Barrett *et al.*, 2006). The primary hosts are goats and sheep, but camels appear to be highly susceptible; and although many wild ungulates have tested positive on serology, a wildlife reservoir has not been identified (Kock *et al.*, 2006).

The epidemiological role of wildlife in PPR circulation is unknown. Serological evidence for infection across a wide spectrum of artiodactyls, sero-conversion of asymptomatic buffalo across multiple ecosystems in Uganda between 2003 and 2004 and PCR evidence for virus presence in healthy buffalo in Côte d'Ivoire (Couacy-Hymann *et al.*, 2005) support a potential wildlife reservoir. The continued southern spread of PPR in African livestock, recent mortalities from PPR among wild ibex in South Asia (personal communication, 2009) and the increasing evidence for serious disease outcomes across a range of small and large wild ungulate species in the Middle East require greater international attention. This disease has the potential to have severe economic impacts and threaten the conservation of biodiversity.

Transmission is by close contact or inhalation of nasal and ocular secretions, saliva, urine, feces, milk and fomites (water and feed troughs and bedding). The virus does not remain viable in the environment for an extended time. The disease is frequently introduced by introducing infected animals into a herd; therefore, quarantine and testing are crucial. Infected and exposed animals are euthanized and cleaning and disinfection of infected premises are recommended (Rossiter, 2004). Vaccination is an effective control measure as immunity persists for at least 4 years and is possibly lifelong. To prevent infection in susceptible populations, it may help to prevent wildlife from having contact with sheep and goats, but more surveillance needs to be done on wildlife to identify their role and their transmission potential. Vaccination might also be possible in wildlife in the near future (Barrett *et al.*, 2006).

Another fatal disease in cattle is malignant catarrhal fever (MCF), caused by the ruminant gamma-herpesviruses alcelaphine herpesvirus 1 (AlHV-1) and ovine herpesvirus 2 (OvHV-2). These viruses cause inapparent infection in their reservoir hosts (wildebeest for AlHV-1 and sheep for OvHV-2), but are fatal, causing fever, inappetence and ocular and nasal discharge when they infect MCF-susceptible species, including cattle, cervids, bison, water buffalo and pigs (Meteyer *et al.*, 1989; Russell *et al.*, 2009). MCF is a significant disease where reservoir and MCF-susceptible species mix and currently is a specific problem in cattle in Bali, Indonesia, bison in the United States and pastoralist cattle herds in eastern and southern Africa. Transmission of both strains of MCF to cattle occurs by contact with placentas at the time of birthing, mainly from wildebeest calves and lambs (Mushi and Rurangirwa, 1981). Control measures are to maintain separation of susceptible animals from infected animals. Cattle are kept separate from wildebeest at calving when the virus is present on pasture from placental contamination. Masai tribesmen delay the movement to pasture until the sun has dried the placenta of the previous night's wildebeest calving (P.B. Rossiter, personal communication 2005). To reduce spill-over into wildlife, bison, deer and other highly susceptible species should not be allowed to be near potentially infected sheep or wildebeest (Russell *et al.*, 2009).

Probably the most ancient viral disease is rabies that is found worldwide. The primary hosts include carnivores, bats and cattle (Swanepoel, 2004). Species-to-species transmission occurs when there is contact with infectious saliva through a bite, mucous membranes or breaks in the skin. All mammals are susceptible to rabies, but reservoir hosts are geographically dependent (Table 1). There are many wildlife vectors, and the disease normally is spread from bites; however, horizontal transmission has been reported in kudu antelope in southern Africa (Mansfield *et al.*, 2006). Respiratory transmission has been documented from bats to other species in enclosed areas, such as caves and houses. In South America, there are significant losses of cattle due to vampire bat bites. To prevent transmission in domestic animals, vaccination, nocturnal housing and the reduction of roaming are useful. Wild European red fox (*Vulpes vulpes*) populations have been successfully immunized with oral bait vaccines. There is some progress in wildlife vaccine use in the United States, but complex carnivore communities prevent its use in some continents like Africa due to difficulties in achieving adequate coverage of target species. Control programs are best targeted at domestic species (Rupprecht *et al.*, 2001).

Bacterial diseases

Brucellosis is spread worldwide and consists of six species, *Brucella abortus*, *Brucella melitensis*, *Brucella*

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suis, *Brucella ovis*, *Brucella canis* and *Brucella neotomae*. The most common species is *B. abortus* that primarily infects bovines causing abortion leading to economic loss and barriers to international trade. The infection is transmitted through contact with fetal membranes, lochia, post-parturient discharges and milk. However, in wildlife little is known about the occurrence of abortions due to brucellosis. *B. abortus* and *B. suis* have been isolated worldwide from a range of wildlife species (Table 2) (Davis, 1990; Pappas, 2010). Only a few true reservoirs can sustain infection with *B. abortus*, bison (*Bison bison*) and elk or wapiti (*Cervis canadensis*) in North America and African buffalo in southern Africa (Godfroid, 2002). In South Africa, hippopotamus (*Hippopotamus amphibious*), zebra (*Equus burchellii*), eland (*Taurotragus oryx*), waterbuck (*Kobus elipsiprymnus*) and impala have tested positive serologically for brucellosis. Apart from *B. abortus* and *B. suis*, the other brucella strains are seldom reported in wildlife. Prevention of wildlife contacting livestock has proven to be effective in controlling the spread of brucellosis in South Africa (Godfroid, 2002). Vaccines are still a work in progress as they do not have the same protection in wildlife but have shown promise in red deer (*Cervus elaphus*) (Godfroid, 2002).

Great economic loss due to chronic disease and death and trade restrictions in cattle can occur with bovine tuberculosis caused by *Mycobacterium bovis* (Pollock and Neill, 2002). Cattle are the primary host but many domestic and wild animals can also vector the disease. Brush-tailed opossums (*Trichosurus vulpecula*), badgers (*Melus melus*), bison, elk, kudu, African buffalo and white-tailed deer (*Odocoileus virginianus*) can all act as reservoir hosts There are numerous spill-over hosts, and species with large populations can become reservoir hosts, e.g. wild boar (Table 2) (Cousins *et al.*, 2004; Corner, 2006). Wild birds appear to be susceptible to *M. bovis* when challenged experimentally (Butler *et al.*, 2001). Currently, there are active elimination programs in the Americas, Japan and New Zealand. The disease is widespread in Africa, Asia and the Middle East (de Lisle *et al.*, 2002) and persists in parts of Europe e.g. United Kingdom, Spain and Portugal. Transmission between animals is by inhalation, ingestion or contamination of an open wound by infected material (respiratory secretions, feces, milk, urine, vaginal secretions or semen) (Cousins *et al.*, 2004; Wilkinson *et al.*, 2009).

Test and slaughtering or segregation can control tuberculosis outbreaks. Affected herds should be re-tested periodically to eliminate shedding in cattle. In the final stages of elimination, surveillance by slaughter and disinfection may reduce the spread of the agent within the herd. The occurrence of *M. bovis* in wildlife reservoirs greatly complicates these efforts. Culling may decrease transmission by decreasing the population density of the reservoir; however, recent findings demonstrated the incidence of cattle tuberculosis was not sustainable once badger culling stopped suggesting that badger culling is not an effective method to control the disease in cattle (Jenkins *et al.*, 2010). Implementation of animal-proof barriers around feeding and storage areas and decreasing interactions between wildlife and domestic animals can decrease transmission. Vaccination of animals against *M. bovis* is key for control where test and slaughter campaigns are not feasible, but its effectiveness in free-ranging wildlife populations is yet to be proven although studies are ongoing (Buddle *et al.*, 2006).

Paratuberculosis, caused by *Mycobacterium avium paratuberculosis* (MAP), causes muscle wasting, decreased milk yield, diarrhea and severe malnutrition leading to death in ruminants which are the primary host. Many infected animals become chronic carriers with only a few developing disease, most commonly presenting as a decrease in milk production. It is spread worldwide except for Sweden and some regions of Australia. In some cases, wildlife can maintain the organism, such as the tule elk (*Cervus elaphus nannodes*) (Manning *et al.*, 2003) and bighorn sheep (*Ovis canadensis*) in the United States (Jessup and Williams, 1999). In The Netherlands and Italy, it is likely that roe deer (*Capreolus capreolus*) play a role in infecting livestock. Similarly, in Scotland, rabbits are thought to have a role in the maintenance and spread of the disease (Daniels *et al.*, 2003a, b). Wild animals found to be infected with the bacterium include a wide variety of small mammals, lagomorphs, bears, primates and pigs (Table 2). MAP bacteria have been detected in birds (*Corvidae*), but their role in transmission is unknown.

The fecal–oral route, fomites and insects are the primary routes of transmission between animals (Buergelt *et al.*, 2004). Wildlife may be able to spread the disease between farms; however, little is known about the transmission in wildlife. Experimental studies and surveys suggest that transmission between ruminants and wildlife can occur (Beard *et al.*, 2001; Corn *et al.*, 2005). Potential routes of transmission from wildlife to livestock could be contamination of pastures, water supplies or feed. There is no treatment so control and/or eradication of the organism are important. For control, wildlife should be kept away from livestock and a test and slaughter program should be implemented for control in cattle (Williams, 2001).

One of the oldest documented bacterial diseases is anthrax, a highly infectious disease of domestic and wild mammals. The disease is common in Eastern Europe, Asia, central and south America and Africa (Beyer and Turnbull, 2009). The disease is caused by *Bacillis anthracis* that has a biotic (amplification phase in mammalian host) and abiotic (dormant spore phase) cycle. When an infected animal dies, sporulation takes place once the vegetative form of the bacterium is exposed to air and spores can remain infectious for decades in the soil. Spores can be disseminated by insects that feed on a carcass and then contaminate browse, by scavengers that feed off the carcass and then contaminate

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Table 1. Infectious vector-borne diseases that are important at the wildlife–livestock interface

Disease (agent)	Vector	Maintenance host[a]	Spill-over host[b]	Reference
Anaplasmosis[c] (*Anaplasma* spp.)	Ticks and flies	White-footed mouse (*Peromyscus leucopus*), raccoon, gray squirrel, reindeer	Cattle, reindeer, horse, dog	Potgieter and Stoltsz (2004).
Anthrax[c] (*Bacillis anthracis*)	Blowflies	Environmental reservoir	Cattle	de Vos and Turnbull (2004)
Bovine petechial fever (*Cytocoetes ondiri*)	Ticks	Bushbuck (*Tragelaphus scriptus*)	Cattle	Sumption and Scott (2004)
Heartwater (*Cowdria ruminantium*)	*Amblyomma* spp.	Buffalo, perissodactyls, artiodactyls, chelonians and gallinaceous bird	Cattle, sheep and wild ruminants	Allsopp *et al.* (2004)
Lyme disease (*Borrelia burgdorferi*)	*Ixodes* spp.	White-footed mouse, wood mouse (*Apodemus sylvaticus*), voles (*Clethrionomys* spp.), dog, white tailed deer, opossum, chipmunk, raccoon, eastern chipmunks, gray squirrels	Dog, horse, cattle, cats; rabbits, mice, hamsters, gerbils and guinea pigs (experimentally)	Reed *et al.* (2003)
Tularemia (*Francisella tularensis*)	Ticks, biting flies, mosquitoes, direct transmission	Lagomorphs (cottontail rabbits, jack rabbits and hares (*Lepus* spp.)), muskrats (*Ondatra zibethicus*), beavers, a variety of rodents including voles (*Microtus* spp.), field mice, vole rats, squirrels and lemmings	Sheep, cats, captive prairie dog (*Cynomys* spp.), farmed mink, dogs, pigs, horses, non-human primates; over 200 species can be affected Infections have been reported occasionally in birds, reptiles, amphibians, crayfish, mollusks and fish	Morner and Addison (2001)
African horse sickness (*Orbivirus*)	*Culicoides* spp., mosquitoes, ticks	Zebra, donkey	Horse, mule, donkey	Coetzer and Guthrie (2004)
African swine fever (*Afsavirus*)	*Ornithodoros* spp. ticks	Wild African suids	Domestic suids	Tulman *et al.* (2009)
Bluetongue (*Orbivirus*)	*Culicoides* spp., mosquitoes	White tailed deer, sheep, cattle, alpacas, llamas	Goat, fallow deer (*Dama dama*), black tailed deer (*Odocoileus hemioneus columbianus*), bison, European red deer, goitered gazelle, mouflon (*Ovis aries musimon*), barbary sheep (*Ammotragus lervia*), Spanish ibex (*Capra pyrenaica*), ibex (*Capra ibex*), southern chamois (*Rupicapra pyrenaica*), musk oxen, yaks, bison, dog, (*Canis lupus familiaris*), Eurasian lynx	Falconi *et al.* (2011)

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Table 1 (Continued)

Disease (agent)	Vector	Maintenance host[a]	Spill-over host[b]	Reference
Epizootic haemorrhagic disease (*Orbivirus*)	*Culicoides* spp.	Cattle, sheep	White-tailed deer, mule deer (*Odocoileus hemionus*), pronghorns (*Antilocapra americana*), Arabian oryx, bighorn sheep, moose, black rhinoceros (*Diceros bicornis*), white rhinoceros (*Ceratotherium simum*), black bears (*Ursus americanus*), panthers (*Felis concolor*)	MacLachlan and Osburn (2004)
Louping ill (*Flavivirus*)	*Ripicephalus* spp., *Ixodes* spp., *Haemaphysalis anatolicum*	Sheep	Cattle, horses, pigs, red deer, dogs, red grouse (*Lagopus lagopus scotticus*), roe deer, feral goats, mountain hare (*Lepus timidus*), short-tailed voles (*Microtus agrestis*), badgers, common shrews (*Sorex araneus*, wood mice (*Apodemus sylvaticus*)	Swanepoel and Laurenson (2004)
Lumpy skin disease (*Poxvirus*)	*Aedes* and *Culex* spp.	Cattle	Asian buffalo (*Bubalus bubalis*), Arabian oryx, springbok (*Antidorcas marsupialis*), oryx, giraffe (*Giraffa cameloparadalis*), impala (*Aepyceros melampus*). Serology only – African buffalo in Kenya and blue wildebeest (*Connochaetes taurinus*), black wildebeest (*Connochaetes gnou*), eland in South Africa	Coetzer (2004)
Japanese encephalitis (*Flavivirus*)	*Culex tritaeniorhynchus*	Wild birds	Cattle, horse and pig	Pfeffer and Dobler (2010)
Tickborne encephalitis[c] (*Flavivirus*)	*Ixodes* spp.	Rodents, large mammals (boar, fox, reindeer and deer)	Goat, sheep, cattle, dog and horse	Mansfield *et al.* (2009)
Western equine encephalitis (*Togovirus*)	*Mosquitoes*	Wild bird and jack rabbit	Horse and emu	Zacks and Paessler (2010)
West Nile disease[c] (*Flavivirus*)	*Mosquitoes*	Wild bird	Horse, donkey and mule	Reed *et al.* (2003)
East coast fever (*Theileria parva lawrenci*)	*Rhipicephalus* spp.	African buffalo	Cattle	Lawrence and Williamson (2004)
Trypanosomosis[c] (*Trypanosoma* spp.)	*Glossina* spp.	African elephant (*Loxodonta* spp.), black rhinoceros, wild bovid and suid	Cattle, horse, pig, sheep, goat and dog	Conner and Van den Bossche (2004)

[a]Usually asymptomatic, reservoir and naturally infected; [b]symptomatic, mortality and possible vectors, [c]Zoonotic disease.

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Table 2. Infectious non-vector-borne diseases that are important at the wildlife–livestock interface

Disease (Agent)	Maintenance host[a]	Spill-over host[b]	Reference
Brucellosis[c] (*Brucella* spp.)	*B. abortus*: cattle, African buffalo (*Syncerus caffer*), bison, elk or wapiti,; *B. melitensis*: sheep, goats, camel; *B. suis*: feral pig (*Sus scrofa*), wild boar (*Sus scrofa*), reindeer, caribou (*Rangifer tarandus groenlandicus*); *B. canis*: canine; *B. ovis*: ram; *B. neotomae*: rodents (*Neotomae lepida*); *B. ceti*: porpoises, dolphins, whale; *B. pinnpedialis*: pinnipeds; *B. microti*: red foxes (*Vulpes vulpes*), common voles (*Microtus arvalis*)	*B. abortus*: yaks, camel, waterbuck (*Kobus elipsiprymnus*), hippopotamus, zebra; *B. melitensis:* chamois (*Rupicapra rupicapra*), ibex (*Capra ibex*), impala; *B. suis*: European hare (*Lepus europaeus*), moose (*Alces alces*)	Bell *et al.* (1976); Davis (1990); Bengis *et al.* (2002); Pappas (2010)
Leptospirosis[c] (*Leptospira* spp.)	Wild mammal and rodents	Cattle, sheep, pig, dog	Leighton and Kuiken (2001)
Paratuberculosis[c] (*Mycobacterium avium* subspp. paratuberculosis)	Cattle (*Bos taurus*), sheep (*Ovis aries*), goats (*Capra hircus*), farmed deer, tule elk, bighorn sheep and roe deer	Armadillo (*Dasypus novemcinctus*), badgers, bear, crows (*Corvus corone*), cat (*Felis catus*), common snipe (*Gallinago gallinago*), cotton rat (*Sigmodon hispidus*), eastern cottontail (*Sylvilagus floridanus*), European brown hares (*Lepus europaeus*), foxes (*Vulpes vulpes*), house sparrow (*Passer domesticus*), jackdaws (*Corvus monedula*), nonhuman primate, Norway rats (*Rattus norvegicus*), opossum (*Didelphis virginiana*), pig, pocket gopher, rabbit (*Oryctolagus cuniculus*), raccoon (*Procyon lotor*), rooks (*Corvus frugilegus*),shrew (*Blarina brevicauda*), sparrow (*Sturnus vulgaris*), stoats (*Mustela erminea*), striped skunk (*Mephitis mephitis*), weasels (*Mustela nivalis*), wood mice (*Apodemus sylvaticus*), white-tailed deer (*Odocoileus virginianus*)	Williams (2001); Corn *et al.* (2005)
Bovine tuberculosis[c] (*Mycobacterium bovis*)	Cattle	Wild bovids, wild suids, felids, opossum, badger, buffalo, bison, elk, kudu, white-tailed deer and primates	Cousins *et al.* (2004)
Avian influenza[c] (*influenza A virus*)	Wild bird (waterfowl and shorebirds)	Wild bird, poultry and mammals	Alexander (2007)
Classical swine fever (*Asfarvirus*)	Wild boar	Swine	Kleiboeker (2002)
Contagious bovine pleuropneumonia (*Mycoplasma* spp.)	Unknown	Cattle	Thiaucourt (2004)
Encephalomyocarditis (*Cardiovirus*)	Rodent	Swine, horse, cattle, primate, mongoose, opossum, raccoon, squirrel, wild birds, African elephant	Grobler *et al.* (1995); Alexander, (2004)
Foot and mouth disease (*Apthovirus*)	Cattle, African buffalo (SAT types)	Cattle, sheep, goat, pig, camel (experimental), water buffalo gazelle, most cloven-footed animals	Thomson and Bastos (2004)

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Table 2 (Continued)

Disease (Agent)	Maintenance host[a]	Spill-over host[b]	Reference
Malignant catarrhal fever (*Alcelaphine herpes virus 1*) (*Ovine herpes virus 2*)	Blue wildebeest (*Connochaetes taurinus taurinus*), white-bearded wildebeest (*C. t. albojubatus*), black wildebeest (*C. gnou*) Sheep	Cattle, pigs, farmed and captive cervids, bison (*Bison bison*), water buffalo (*Bubalus bubalis*); captive eland (*Taurotragus derbianus*), sitatunga (*Tragelaphus spekei*) and giraffe (*Giraffa camelopardalis*); antibodies in oryx spp., roan (Hippotragus equines), sable (hippotragus niger), musk ox (Ovibos moschatus), blesbok (Damaliscus dorcas phillipsi), topi (Damaliscus korrigum), red hartebeest, *Alcelaphus buselaphus*), *Bali cattle in Indonesia* (*Box javanicus*), *Pere David's deer* (*Elaphurus davidianus*), *African buffalo*; experimentally rabbits and Syrian hamsters	Reid and Van Vuuren (2004); Russell *et al.* (2009)
Newcastle disease[c] (*Avulavirus*)	Wild and domestic birds particularly cormorants and pigeons	Poultry	Alexander (2000)
Peste des petits ruminants (*Morbillivirus*)	Domestic small livestock	Wild goat, sheep, camel, wild ibex, semi-captive gazelle, oryx, other wild ungulates	Rossiter (2004)
Rabies[c] (*Lyssa virus*)	North America: many species of insectivorous and vampire bats but most commonly silver-haired bat (*Lasionycteris noctivagans*), eastern pipistrelle (*Pipistrellus subfavus*), striped skunk, raccoon, red fox (*Vulpes vulpes*), grey foxes (*Urocyon cinereoargenteus*), coyotes (*Canis latrans*); Central and South America: 3 species of vampire bats, most commonly transmitted by common vampire bat (*Desmodus rotundus*); Caribbean countries: grey mongoose (*Herpestes auropunctatus*); Europe, Greenland, Russia: red fox; Africa, Asia, Central and South America: dog; Asia, eastern Europe: wolf (*Canis lupus*); Africa: yellow mongoose (*Cynictis penicillata*), black-backed jackal (*Canis mesomelas*), side-striped (*Canis adustus*), bat eared fox (*Otocyon megalotis*); Russia: raccoon dog (*Urocyon cinereoargenteus*)	Dog, cat and cattle	Swanepoel (2004)
Rinderpest (*Morbillivirus*)	Cattle, wild ruminants (?)	Cattle, sheep, goat, yak, wild African ruminants and suids, wild and domestic cloven-hoofed animals	Roeder and Taylor (2002)
Fascioliosis (*Fasciola hepatica*) (*F. gigantica*)	IH: fresh water snail (*Galba truncatula*) in Europe, parts of Asia DH: sheep and cattle primarily IH: fresh water snail (*Lymnaea natalensis*) in Africa and (*Lymnaea rubiginosa*) in Asia and vegetation DH: cattle and water buffalo	Coypu (*Myocaster coypus*), capybara (*Hydrochoerus capybara*), llama (*Lama glama*), alpaca (*Lama pacos*), vicunas (*Vicugna vicugna*), guanacos (*Lama guanicoe*), elk (*Alces alces*), deer (*Capreolus capreolus*), beaver (*Castor fiber*), red deer (*Cervus elaphus*), hare (*Lepus europaeus*), otter (*Lutra lutra*), wild boar (*Sus scrofa*), kangaroo, camel, rabbits, goats, horse	Shimalov and Shimaolv (2000); Robinson and Dalton (2009); Pietrokovsky *et al.* (2009)
Hydatid disease[c] (*Echinococcus multiocularis*)	IH: rodents, domestic and wild pig, dog, monkey, horse?; DH: fox, dog, cat, wolf, raccoon dog, coyote	Unknown	Jenkins *et al.* (2005)

Table 2 (Continued)

Disease (Agent)	Maintenance host[a]	Spill-over host[b]	Reference
(*Echinococcus granularis*) Cervid/G8 and G10	IH: cervids; DH: wolf, dog		
Sheep/G1	IH: sheep, cattle, pigs, camels, goats, macropods; DH: dog, fox, dingo, jackal and hyena		
Tasmanian sheep/G2	IH: sheep, cattle?; DH: dog, fox		
Buffalo/G2	IH: buffalo, cattle?; DH: dog, fox?		
Nipah virus encephalitis[c]	Fruit bats (*Pteropus* spp.)	Pigs, horses, dogs, cats	Lo and Rota (2008)

[a]Asymptomatic and reservoir; [b]symptomatic, mortality and possible vectors, [c]Zoonotic disease; DH, definitive host, IH, intermediate host.

water supplies, by water run-off contaminating pastures and directly by ingesting the carcass.

Herbivores and suids are the primary hosts. Death due to anthrax has been recorded in at least 52 species (Hugh-Jones and de Vos, 2002) including many antelope hunted for human meat consumption. Oftentimes, wildlife epizootics follow severe outbreaks in livestock; they also occur when there is a high infection load in the environment and at the wildlife–livestock interface when land and resources are being shared by both. However, outbreaks in cattle have occurred in the absence of any wildlife and vice versa (Beyer and Turnbull, 2009). Anthrax once introduced into wildlife populations remains one of the most devastating diseases shown by recently reported outbreaks affecting wildlife in the Americas, Africa and Asia (Mlengeya *et al.*, 1998; Manyibe *et al.*, 2006; Clegg *et al.*, 2007; ProMed-mail, 2010a).

Cattle and wildlife can be protected by vaccination, but in wildlife there are few examples where it has been applied early in an epidemic. One apparently successful example of vaccine use was during a recent outbreak among endangered Grevy's zebra (*Equus grevyi*) in northern Kenya (Muoria *et al.*, 2009), where both cattle and zebra were vaccinated but it is extremely difficult to scientifically assess the efficacy and impact on the epidemic. Other means of control are burning, burying or plastic covering of infected carcasses, preventing scavenging, disinfecting contaminated sites and controlling watering sites at which scavengers bathe by replacing them with concrete water troughs (Hugh-Jones and de Vos, 2002). Development of an oral vaccine for wildlife is warranted.

Leptospira has over 200 serovars that belong to 10 species (Murray, 2003). Leptospirosis is becoming more common in tropical areas but is not exclusive to hot climates. Increased prevalence of this zoonosis has been associated with rising human and livestock populations. The rising populations lead to increases in contamination and deterioration of water sources as well as adaptation of certain wildlife species to degraded human landscapes, thriving on human waste and detritus (Vinetz *et al.* 2005). The organism is excreted in the urine and through secretions associated with calving, and it can be transmitted directly or indirectly between hosts (Hunter, 2004). All mammals are susceptible to at least one species of *Leptospira*, and this pathogen has been identified in over 180 species of animals.

The primary reservoir hosts for most species of *Leptospira* are wild mammals, particularly rodents, and domestic hosts are cattle, pigs, sheep and dogs (Hunter, 2004). Rats are a reservoir for *Leptospira icterohaemorrhagiae* that affects cows, pigs and mice and are also a reservoir for *Leptospira ballum* which causes disease in sheep (Leighton and Kuiken, 2001). The most prominent signs in affected sheep and cattle are abortions, decreased fertility or decreased milk yield and in swine, stillbirths and increased neonatal mortality (Hunter, 2004). Infection

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in reservoirs is most commonly asymptomatic (Leighton and Kuiken, 2001) but wildlife can be adversely affected, with certain marine mammal species being highly susceptible and therefore good indicators of estuary pollution (Vinetz *et al.* 2005). For control in pigs, cattle and dogs, there are vaccines; however, vaccines prevent symptoms, not necessarily infection or shedding of organisms (Hunter, 2004). Immunity is serovar specific and vaccines are not cross-protective. Preventing contact with infected wildlife, livestock or contaminated environments and controlling rodent populations and good sanitation in kennels and areas where livestock are bred or give birth will reduce occurrence of the disease. Longer-term solutions to reduce the risk of exposure require restoration of ecological services and improving agricultural systems or practices that reduce contaminated run-off entering waterways (Vinetz *et al.* 2005).

Porcine viral diseases

African swine fever virus (ASFV) is highly contagious and spreads rapidly in susceptible pig populations (Fauquet *et al.*, 2005). Isolates of ASFV vary in terms of virulence from being highly pathogenic, causing nearly 100% mortality, to low-virulence isolates that can be difficult to diagnose. The disease is endemic in most of sub-Saharan Africa, the Caucasus and the island of Sardinia, Italy. The primary hosts are domestic and wild boar, which are equally susceptible to the disease. Warthog (*Phacochoerus africanus*) are the natural host in sub-Saharan Africa, and they tolerate infection without clinical disease (Tulman *et al.*, 2009). In the presence of *Ornithodoros* spp. ticks, the cycle is maintained through the infection of piglets in the burrow. Maintenance in domestic pigs in the absence of *Ornithodoros* spp. ticks or reservoirs is dependent on susceptible hosts (pigs and wild boar) in high density and with high reproductive or turnover rates to ensure the presence of naïve hosts. Free-ranging village pigs and feral pigs are important reservoirs of infection in many parts of western Africa. However, ASFV-infected products often act as a reservoir, causing the disease to reappear when infected products are fed to pigs. For example, an outbreak of ASFV was recorded in pigs after being fed guinea fowl that had eaten ASFV-infected ticks (Kleiboeker, 2002).

Although not fully understood, the role wild boar may play in the introduction, persistence and spread of the virus in Eurasia must be further investigated. Whether wild boar are spill-over hosts or whether they can maintain the disease is still in debate. Transmission occurs by direct contact with infected animals through the oro-nasal route, indirect contact on fomites (vehicles, feed and equipments), aerosol transmission (only over very short distances) and by the bite of infected *Ornithodoros* spp. ticks. For control measures in endemic areas, it is difficult to eliminate the natural reservoir; however, control of the soft tick vectors, regulation and/or banning swill feeding and adoption of high biosecurity when infection occurs are important in preventing and controlling the disease (Tulman *et al.*, 2009).

Classical swine fever (CSF) is another highly contagious and economically significant viral disease of pigs (Fauquet *et al.*, 2005). There have been outbreaks of CSF in the United Kingdom, Netherlands, Romania, Slovakia, Spain and Germany; although, many countries have eliminated this disease from domestic swine (Stegeman *et al.*, 2000; Artois *et al.*, 2002). The primary hosts are the same as for ASFV, domestic pigs; however, feral pigs are thought to be susceptible to infection and are therefore potential reservoirs. Although it is still debated, some believe that wild pigs cannot sustain an infection without the presence of CSF virus (CSFV)-infected domestic pigs (Kleiboeker, 2002). In Sardinia, however, it appears that wild boar-to-wild boar transmission played an important role in virus persistence; in contrast, ASFV was unable to persist in the absence of free-roaming domestic swine (Artois *et al.*, 2002). Transmission of CSFV occurs by direct contact or ingesting infected food through unregulated swill feeding. For controlling the disease in affected areas, the following are necessary: quarantine, banning wild boar hunting, oral vaccination of wild boars, culling and testing of wild boars and vaccination of domestic pigs in the region affected (Artois *et al.*, 2002; Kleiboeker, 2002).

Nipah virus infection is emerging in Southeast Asia. The virus is carried in fruit bats (reservoir) and affects humans and pigs (hosts). The virus began circulating in pigs in Malaysia in 1996 and then emerged in humans in 1998 when over 260 people became ill and 1 million pigs had to be culled. Nipah virus is now endemic in Southeast Asia. Swine are extremely susceptible and can act as an amplifying host for the virus. The primary reservoirs are fruit bats of the genus *Pteropus*. Transmission is unknown but is considered to be from eating fruit that has been contaminated with bat saliva or urine, by drinking contaminated water or by eating aborted bat fetuses. Where close contact with infected pigs has occurred, infections have been reported in dogs, horses and cats (Daniels *et al.*, 2004). For control, contact between pigs and fruit bats needs to be decreased. Fruit tree plantations should be removed from areas where pigs are kept, wire screens should be placed on open-sided pig sheds and run-off from the roof should be prevented from entering pens where pigs reside. Animal models used for vaccine development appear promising (Lo and Rota, 2008).

Avian viral diseases

Newcastle disease virus (NDV) is caused by avian paramyxovirus type 1 and is an important viral disease of domestic birds causing morbidity and mortality rates up to 100% in poultry (Alexander, 2000). Poultry are the primary host, but over 250 species of birds in 27 orders

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have been reported infected (Alexander, 2000). Wild birds can act as a reservoir and large die-offs have occurred as a result of ND virus (NDV). For example, there was 92% mortality in juvenile cormorant populations in Canada and the United States (Glaser *et al.*, 1999). *Columbiformes* are also susceptible to NDV (ProMed-mail, 2009a), and NDV is suspected of causing large mortalities among pigeons and doves in East Africa over the last decade (R.A. Kock, personal observation 2006). Wild birds can carry the virus that becomes more virulent once established in poultry.

A strain of virus found in cormorants has been detected in domestic turkeys, although the route of transmission between the wild and domestic birds is unclear (Kuiken, 1999). Recently a strain of NDV has been confirmed in another large-scale die-off of double-crested cormorants in the United States (ProMed-mail, 2010b). Transmission is through inhalation or ingestion of the virus in feces and respiratory secretions or by fomites (Kuiken, 1999). To reduce transmission of NDV, good poultry biosecurity is necessary, which includes bird and pest proofing of housing, clean feed and water supplies, minimizing travel on and off the facility and disinfecting vehicles and equipment before and after they enter the farm (Alexander, 2000).

Avian influenza viruses are contagious and extremely variable viruses that are widespread in birds, particularly waterfowl and shorebirds (Webster *et al.*, 1992). Epidemics of highly pathogenic avian influenza can spread rapidly, and devastate the poultry industry, resulting in high mortality and severe trade restrictions (Alexander, 2007). Avian influenza is caused by viruses in the influenza virus A genus of the family *Orthomyxoviridae* that are closely related to human, equine, swine and canine influenza viruses. There are two surface antigens, the hemagglutininin (HA) and neuraminidase (NA) proteins, which are used to name the type of influenza A, and the combination of 16 HA and 9 NA proteins determines the subtype (Webster *et al.*, 1992).

Low-pathogenicity avian influenza (LPAI) and highly pathogenic avian influenza (HPAI) viruses are categorized according to the severity of disease in chickens. The vast majority of viruses found in birds are LPAI; however, HPAI viruses are predominately a cause of disease of poultry where intensive rearing conditions appear to encourage some strains of virus to evolve into a pathogenic form (Alexander, 2007). These viruses rarely spill over into wild birds but recent wild bird outbreaks indicate a possibly changing epidemiology with H5N1 having been reported causing wild bird epidemics in China (Liu *et al.*, 2005), Mongolia and Russia in 2005, 2006, 2009 and 2010; but these appear to be self-limiting and persistence has not been demonstrated. These epidemics are likely dependent on continued spill-over from a high infection load in the environment, associated with domestic poultry production systems and especially, domestic and semi-domestic waterfowl (Newman *et al.*, 2010).

Waterfowl and shorebirds appear to be the natural reservoirs of LPAI and carry all the known avian influenza subtypes. Transmission of the virus is by contact with infected feces, saliva and nasal secretions (Alexander, 2007). Fomites are important in the transmission of both LPAI and HPAI and flies may be mechanical vectors. In countries where HPAI has been eliminated from domesticated poultry, potential routes of introduction into flocks are infected migratory waterfowl, poultry, pet birds or fomites (Alexander, 2007), but the wild bird route of infection remains largely unconfirmed and anecdotal (Newman *et al.*, 2010). A wild bird reservoir for H5N1 is yet to be established and possibly never will be, given the widespread infection of wild birds with mild strains of viruses and considerable resilience to H5N1 infection through cross protection. The quite remarkable expression of H5N1 pathogenicity in some wild bird populations is cause for alarm, however, and shows a considerable shift in character of these viruses through co-evolution with poultry.

Control of the disease in poultry is based on preventing the introduction of infected birds, contact with contaminated fomites and contact with wild birds and their water sources. Birds should not be returned to the farm from live bird markets and strict hygiene and biosecurity measures are necessary to prevent virus transmission on fomites. Outbreaks can be controlled by rapid depopulation of infected and exposed flocks, proper disposal of carcasses and contaminated materials and strict biosecurity measures. Vaccines are not often used; however, countries or regions may consider vaccination as a preventive or adjunct control measure during an outbreak (Capua and Marangon, 2006, 2007). Ultimately, control of this disease will most likely require a re-examination of poultry production systems that appear to be a driver for the evolution of highly pathogenic influenza viruses (Wallace *et al.*, 2010).

Future research needs

There are some diseases where the role of wildlife is still open to question. Contagious bovine pleuropneumonia, a very important disease of cattle in Africa, can cause 80% mortality in naïve herds, with the majority of the surviving cattle remaining chronic carriers of the causative bacterium. The last large outbreak was in Zambia in 2006 and approximately 15,000 cattle were culled (ProMed-mail, 2009b). To date, there is no evidence indicating that wildlife is a reservoir for this disease; however, few investigations have been completed to disprove the potential (Barber *et al.*, 1970; Paling *et al.*, 1988). Lumpy skin disease is another example where there are too few investigations to delineate the role of wildlife in the epidemiology of this disease. There have been only a few documented cases in Asian water buffalo (*Bubalus bubalis*) (Ali *et al.*, 1990) and Arabian oryx (*Oryx*

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leucoryx) (Greth *et al.*, 1992). Rift valley fever (RVF), a vector-borne viral disease that causes high mortality rates in young domestic ruminants and abortion in adults, is another example where the role of wildlife needs be determined. Increased surveillance and research on the disease ecology of pathogens will help to define and clarify the role of wildlife in these diseases.

Vector-borne diseases

Vector-borne diseases have the potential to become more prevalent as they are affected by climate change. Although there are many vector-borne diseases that have a wildlife reservoir and are economically important in livestock such as RVF, Japanese encephalitis in swine and trypanosomiasis in cattle, these diseases are not considered contagious because spread of the disease is via insect vectors (*Culex* spp. and tsetse fly, respectively). Therefore, rapid transboundary spread does not usually occur. Another vector species for which climate patterns impact populations and disease prevalence are rodents, which are a reservoir for the agent of encephalomyocarditis. This disease of rodents in which there are spill-over infections to pigs and free-ranging elephants occurs when there are peaks in the rodent population which are driven by rainfall (Grobler *et al.*, 1995; Bengis *et al.*, 2002). For prevention and control of insect vector-borne diseases, combinations of aerial spraying, adhesive pyrethroid insecticides, impregnated screens and traps, vaccines if available, movement of animals to higher altitudes and confining animals to insect-proof stables hold the greatest promise of recovering infested areas for animal production. Rodent vectors can be controlled through trapping, poisoning and rodent-proofing livestock housing and feed stores.

Parasitic diseases such as hydatid disease and fascioliasis, vector-borne and parasitic diseases are also not covered in this paper but are included in the tables that follow (Tables 1 and 2). From a production viewpoint, parasites and tick-borne diseases are particularly important because, while rarely fatal, they can seriously affect productivity and profitability. Although parasites can be effectively controlled, parasite resistance to drugs, arising from inappropriate use, is a growing problem. For ticks, the cost of traditional dipping with acaracides for control is becoming prohibitive, and raises environmental concerns about disposal of the waste chemical (Williams *et al.*, 2002).

Conclusion

Although this paper only summarizes the important TADs at the wildlife–livestock interface, it emphasizes that there is an extensive list of diseases with the potential to be exchanged at this interface and many of the known diseases have unidentified reservoirs and hosts (Tables 1 and 2). Globally, novel diseases are being discovered and many pathogens are re-emerging or moving into new hosts and environments. Human population growth is predicted to reach 9 billion within a few decades, leading to increased encroachment into wildlife-occupied regions for settlement and food production. Furthermore, as urban areas expand and income levels rise in emerging economies, the demand for livestock-based protein will result in increases in the number of domestic animals raised in these areas. This will cause increased contact between the sectors and therefore create more opportunities for pathogen migration and mutation. It is crucial to learn more about the complex ecology of these diseases, identify the reservoirs and vectors for pathogens and identify new and changing epidemiological patterns to reduce future disease transmission.

There is also a need to initiate regional training for capacity building to improve detection of disease in wildlife and domestic livestock, and enhance surveillance to facilitate early detection of disease. It is critical that veterinary and wildlife health professionals work across disciplines to share information about outbreaks for both disease monitoring and control. Specifically, this approach needs to be focused in regions where disease is more likely to emerge, such as tropical countries with growing populations where encroachment into previously 'wild' areas is creating closer contact between domestic and wild animals. These countries include India, China, the Democratic Republic of Congo, Bangladesh, Indonesia, Pakistan and Ethiopia where, according to a 2008 report by FAO, 65% of the world's hungry live (FAO, 2010). When considering the effects of disease on livestock and livelihoods, future studies and surveillance efforts need to be focused on the diseases that have the greatest impact on agriculture, trade and food security in these countries.

Understanding wildlife and their role is a vital part of understanding the epidemiology and ecology of transborder infectious diseases and is necessary for effective control of these diseases. The role of agroecological systems providing conducive environments for pathogen evolution needs to be examined, in light of increasing emergence of diseases in livestock, some with possible origins in wildlife species, based on phylogenetic associations of strains and types of agent. Subsequent spill-back into the environment and to humans and other animals from livestock is also of considerable concern. In order to improve this understanding, the following components are necessary: capacity building and training, cross-sectoral collaboration and information sharing between health professionals, increased wildlife disease surveillance and monitoring, wildlife–livestock interface and disease ecology studies that include habitat use, migration ecology, population monitoring methods and a collaborative approach outbreak investigation. Cross-sectoral cooperation and a multi-faceted One Health

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approach are crucial to clarify disease dynamics at the wildlife and livestock interface and to address important pathogens impacting agricultural livelihoods and conservation of valuable wildlife resources globally.

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